

Distances parcourues, distribution

Cantien Collinet

Mercredi 17 juin 2020

Préparation des données —

Installation des librairies et importation des données

```
source("support/preparation.R")

# Calcul des distances parcourues
distance_brut <- déplacements %>%
  group_by(NQUEST, NP) %>%
  summarise(distance = sum(DPORTEE)) %>%
  right_join(individus_CS8_short, by = c("NQUEST", "NP")) %>%
  mutate(distance = tidyr::replace_na(distance, 0)) %>% # 0 pour les immobiles
  filter(!is.na(CS8)) %>%
  var_labels(distance = "Distance parcourue quotidiennement par personne (en km)")

# Tableau avec les individus de toutes les PCS de niveau 1
distance <- distance_brut %>%
  mutate(CS8 = sjlabelled::as_label(CS8, prefix = TRUE))
distance <- sjlabelled::copy_labels(distance, distance_brut)

# Données pondérées
distance_w <- svydesign(ids = ~1,
  data = distance,
  weights = ~POIDSP)

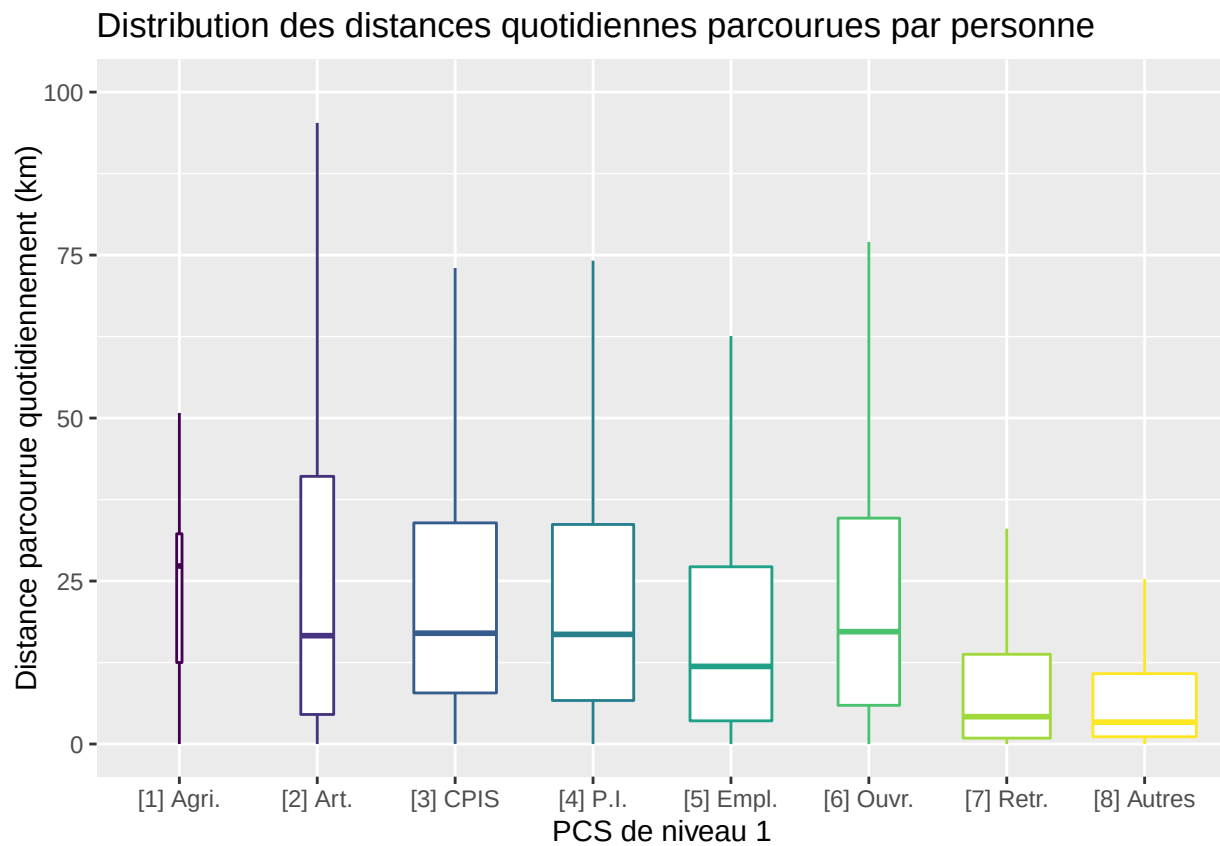
# Tableau avec uniquement les CPIS, employés et ouvriers
distance_subset <- subset(distance_brut, CS8 %in% c(3, 5, 6))
distance_subset <- distance_subset %>%
  mutate(CS8 = sjlabelled::as_label(CS8, prefix = TRUE))
distance_subset <- sjlabelled::copy_labels(distance_subset, distance_brut)

# Données pondérées
distance_subset_w <- svydesign(ids = ~1,
  data = distance_subset,
  weights = ~POIDSP)
```

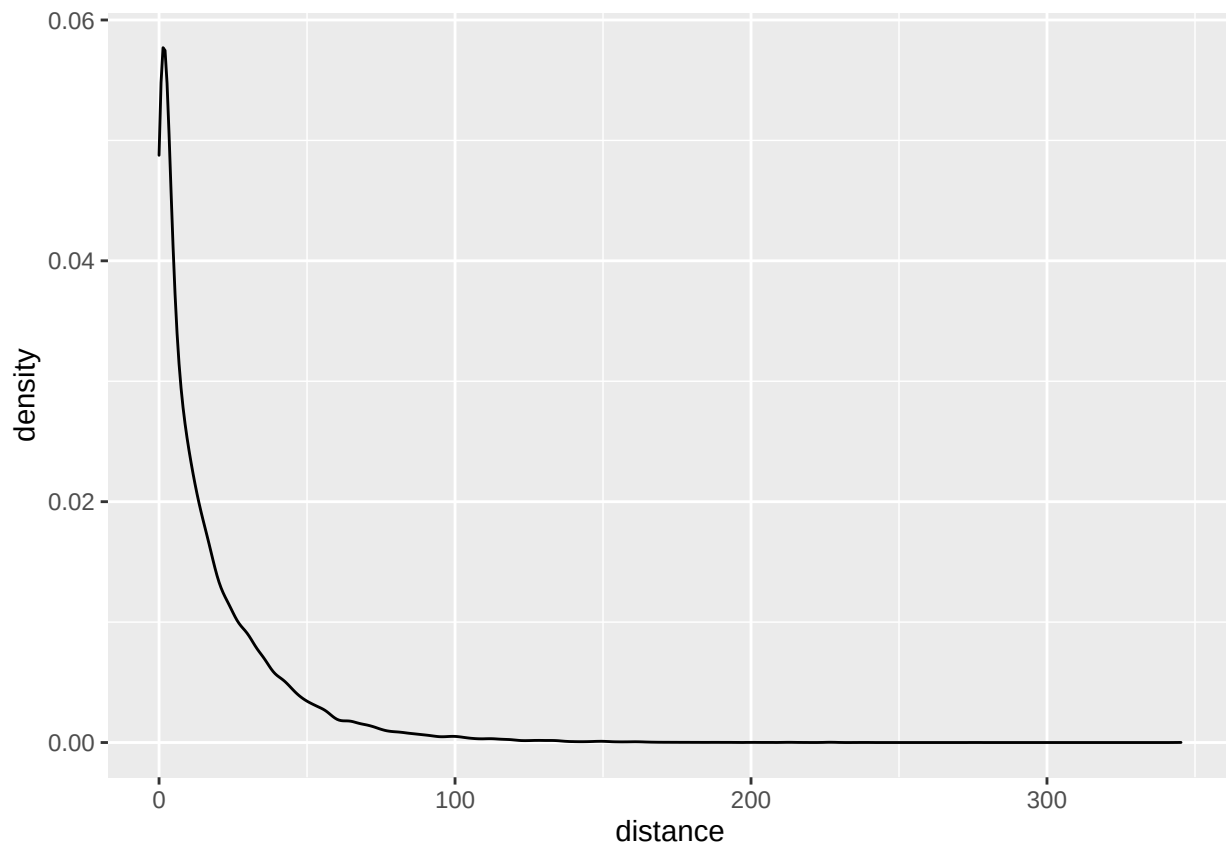
Exploration —

Distribution des temps de déplacements par jour et par personne —

```
ggplot(distance_w$variables, aes(
  weight = weights(distance_w),
  x = CS8,
  y = distance,
  color = CS8)
) +
  geom_boxplot(outlier.shape = NA, varwidth = TRUE) +
  ggtitle("Distribution des distances quotidiennes parcourues par personne") +
  ylab("Distance parcourue quotidiennement (km)") +
  xlab("PCS de niveau 1") +
  guides(color = FALSE) +
  theme(axis.title.y = element_text(vjust = 0)) +
  coord_cartesian(ylim = c(0, 100))
```



```
ggplot(distance_w$variables, aes(weight = weights(distance_w), x = distance)) +
  geom_density()
```



```
distance_moyennes <- svyby(~distance, ~CS8, distance_w, svymean)
```

```
distance_quantiles <- svyby(
  ~distance,
  ~CS8,
  distance_w,
  svyquantile,
  quantiles = c(0.1, 0.25, 0.5, 0.75, 0.9),
  ci = TRUE,
  vartype= "ci"
)
```

```
library(convey)
distance_cw <- convey_prep(distance_w)
distance_gini <- svyby(~distance, ~CS8, distance_cw, svygini)
```

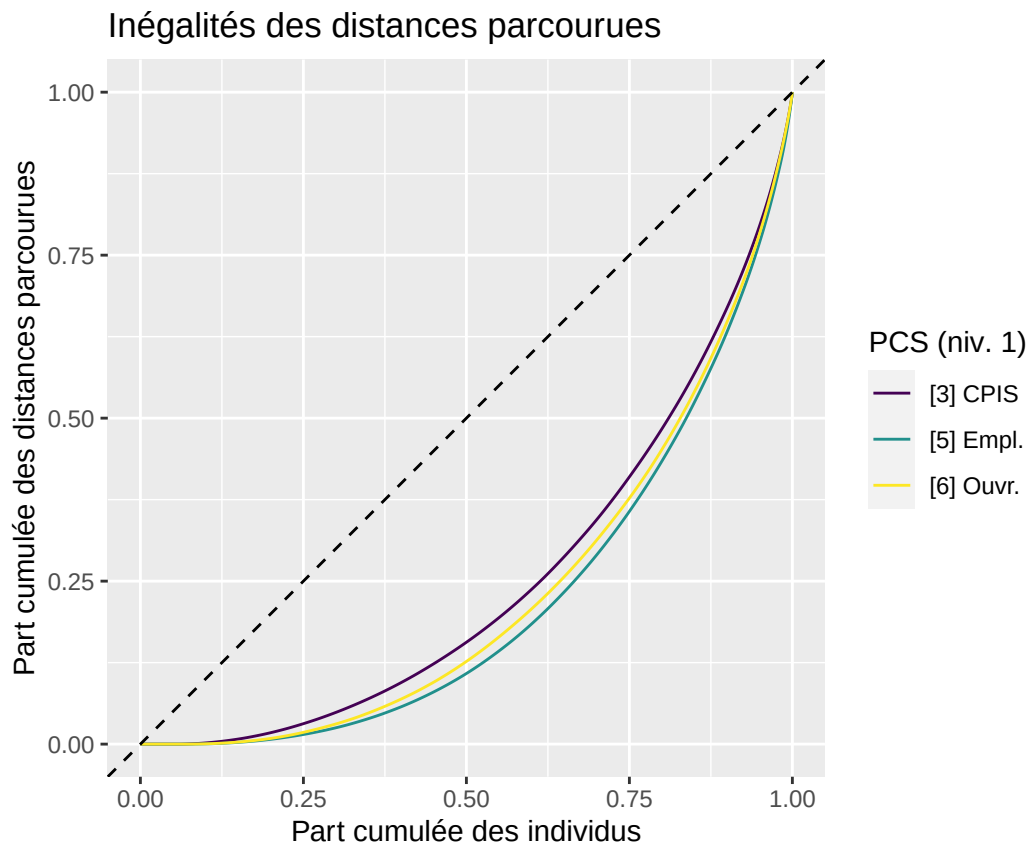
Table 1: Distances quotidiennes parcourues par personne (en km)

PCS (niv. 1)	Moyenne	1er décile	1er quartile	Médiane	3e quartile	9e décile	Coefficient de Gini
[1] Agri.	26.5	0.0	7.8	27.1	31.3	48.3	0.514
[2] Art.	27.8	0.4	4.5	16.6	40.9	74.3	0.578
[3] CPIS	24.1	2.0	7.8	17.0	33.9	54.0	0.494
[4] P.I.	24.4	1.6	6.7	16.8	33.7	56.4	0.513
[5] Empl.	19.0	0.4	3.6	11.9	27.2	45.1	0.561
[6] Ouvr.	25.2	0.8	5.9	17.2	34.6	59.5	0.536
[7] Retr.	10.9	0.0	0.9	4.2	13.8	29.8	0.664
[8] Autres	9.2	0.4	1.1	3.4	10.8	25.7	0.663

```
library(gglorenz)

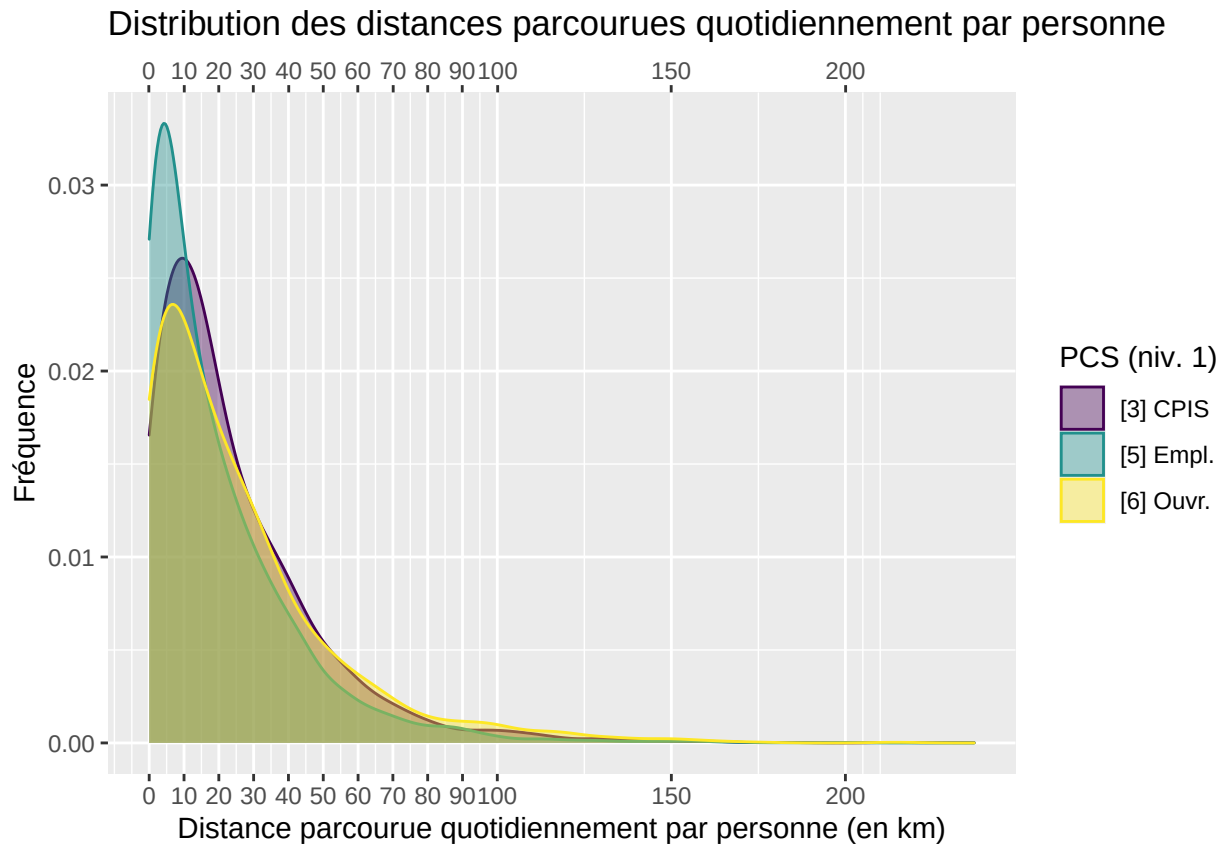
ggplot(distance_subset_w$variables) +
  stat_lorenz(aes(weight = weights(distance_subset_w), x = distance, colour = CS8), desc = FALSE) +
  coord_fixed() +
  geom_abline(linetype = "dashed") +
  scale_colour_discrete(name = sjlabelled::get_label(distance_subset$CS8)) +
  labs(x = "Part cumulée des individus",
       y = "Part cumulée des distances parcourues",
       title = "Inégalités des distances parcourues")
```

Warning: Ignoring unknown aesthetics: weight



```
ggplot(distance_subset_w$variables, aes(
  weight = weights(distance_subset_w),
  x = distance,
  color = CS8,
  fill = CS8
)) +
  geom_density(adjust=1.5, alpha = .4) +
  scale_colour_discrete(name = sjlabelled::get_label(distance_subset$CS8)) +
  scale_fill_discrete(name = sjlabelled::get_label(distance_subset$CS8)) +
  scale_x_continuous(
    breaks = c(seq(0, 100, 10), 150, 200),
    minor_breaks = waiver(),
    sec.axis = dup_axis(name = NULL)
  ) +
```

```
labs(
  title = "Distribution des distances parcourues quotidiennement par personne",
  x = sjlabelled::get_label(distance_subset$distance),
  y = "Fréquence"
)
```



```
ggplot(distance_subset_w$variables, aes(
  weight = weights(distance_subset_w),
  x = distance,
  fill = CS8
)) +
  geom_density(adjust=1.5, position = "fill", alpha = .4) +
  scale_colour_discrete(name = sjlabelled::get_label(distance_subset$CS8)) +
  scale_fill_discrete(name = sjlabelled::get_label(distance_subset$CS8)) +
  scale_x_continuous(
    breaks = c(seq(0, 100, 10), 150, 200),
    minor_breaks = waiver(),
    sec.axis = dup_axis(name = NULL)
  ) +
  scale_y_continuous(
    breaks = c(0, 0.33, 0.66, 1),
    minor_breaks = waiver(),
    sec.axis = dup_axis(name = NULL)
  ) +
  labs(
    title = "Distribution relative des distances parcourues quotidiennement par personne",
    x = sjlabelled::get_label(distance_subset$distance),
```

```
y = "Fréquence relative"  
)
```

